

Termination and confluence of the cherry–bundle reduction: the global assembly (R7) for the Laplacian–ratio maximizer

Abstract

We record the global-assembly step (R7) of the Brualdi–Goldwasser Laplacian–ratio maximizer program as a *terminating, confluent rewriting system* on trees. Given the local optimality results already established — $\Phi \leq 1$ (R3, no open mathematical idea remaining), branch multiplicativity (proved in the $z_H \rightarrow 0$ limit), the Kelmans compression monotonicity (R4/thm:kelmans), the single–hub tiebreak (R5), and the cherry distribution (R6) — we define four amplitude–non–decreasing rewrite rules and prove that they (i) terminate, by a well–founded measure, and (ii) are locally confluent, so by Newman’s lemma the system is confluent with a *unique normal form*. The normal form of every n –vertex tree is the de–loaded single–hub five–cherry–bundle star, which therefore maximizes $\pi(T) = \text{per } L(T) / \prod \deg$ for all large n . [Updated 2026–08–15.] The boundary caveat of earlier drafts (rule monotonicity “verified to $n = 240$ ”) is now closed: the merge layer is proved by exact rational certificates for all sizes — the loaded–donor exchange dichotomy, the topped–up merge (36 load cells, any capped environment), and the stuck–configuration dominations — and is *machine–checked* in Lean 4 (the R47 development: the 36–cell table, head–merge monotonicity, and the capstone `chain_to_normalForm`, which rewrites every certified state monotonically in $\text{per } L / \prod \deg$ to an ordered–merge normal form). The remaining honesty ledger is explicit and short (see §??).

1 Setup and the reduction system

Let T be an n –vertex tree and $\pi(T) = \text{per } L(T) / \prod_v \deg(v)$ its Laplacian ratio. Write $A(T) = \pi(T) / \rho_B^n$ for the *arm–normalized amplitude*, where $\rho_B = (621/64)^{1/11}$ is the branching growth rate. The target (R7’) is

$$A(T) \leq C_1 = \frac{26/23}{\rho_B} \quad \text{for every tree, with equality iff } T = S^*, \quad (1)$$

where S^* is the de–loaded single–hub five–cherry–bundle star (one hub carrying k arms, each arm a center with five length–two cherries, the hub itself de–loaded to zero cherries).

We use four rewrite rules $\rightsquigarrow$, each a local surgery on T . Each is justified *monotone* — A does not decrease — by a previously established result.

Definition 1 (the rules).

- (L) **Leg→cherry.** Replace any leg (a maximal path hanging off a center) that is not a length–two cherry by a length–two cherry attached to the same center. Monotone by R2 (prop:cherries): non–cherry legs are rate–suboptimal, so the move does not decrease A ; strictly increases it unless the leg was already a cherry.
- (B) **Branch→arm.** At a hub, replace any rooted branch G (not a five–cherry arm) by a five–cherry arm. Monotone by R3 ($\Phi \leq 1$) and branch multiplicativity: $A(\text{hub}+G) = A_\star \Phi(G)$ with $\Phi(G) \leq 1$, so replacing G by an arm (where $\Phi = 1$) does not decrease A ; strict unless G was already an arm.

arms compression. Apply a Kelmans transformation moving pendant structure from a lower center toward a higher one along the backbone. Monotone by R4 (thm:kelmans): the compression does not decrease π , hence not A (same n); strict unless the two centers are already coincident/ordered.

merge and de-load. Merge two hubs of a multi-hub backbone into one and rebalance arm counts to five with the hub de-loaded. Monotone by R5 (rem:tie: $C_1^m < C_1$ for $m \geq 2$, exact crux $(26/23)^{11} < 621/64$) and R6 (distribution.py/hub.py: arms balance at five, hub de-loads).

Remark 2. Each rule preserves the vertex count n except (L) and (B), which preserve it when the replaced object has the same size and otherwise are applied at fixed n by the amplitude comparison (the amplitude A , not π , is the monotone quantity, and A is defined at each n). All four are monotone in A ; this is the content imported from R2–R6.

2 Termination

Definition 3 (measure). For a tree T define $\mu(T) = (h(T), b(T), d(T), e(T)) \in \mathbb{N}^4$ ordered lexicographically, where

- $h(T)$ = number of hubs (backbone centers of unbounded-degree type) $- 1$, i.e. the excess over a single hub;
- $b(T)$ = number of branches at hubs that are not five-cherry arms;
- $d(T)$ = total “Kelmans displacement”: $\sum$ over pendant units of their backbone distance from the top hub;
- $e(T)$ = number of legs that are not length-two cherries, plus the deviation $\sum_{\text{arms}} |c_i - 5|$ of arm cherry counts from five, plus the hub cherry load.

Proposition 4 (termination). Every rewrite step strictly decreases μ in the lexicographic order $<$ on $\mathbb{N}^4$. Hence no infinite reduction sequence exists.

Proof. Rule **(H)** strictly decreases h (one fewer hub) and cannot increase it; so μ drops in the first coordinate. Rule **(B)** fixes h (it does not create or remove hubs) and strictly decreases b (one fewer non-arm branch), while it may only decrease e ; so μ drops in the second coordinate with the first fixed. Rule **(K)** fixes h and b (compression neither changes the hub count nor turns a non-arm branch into an arm or vice versa — it moves whole pendant units) and strictly decreases the displacement d ; so μ drops in the third coordinate with the first two fixed. Rule **(L)** and the rebalancing part of the moves fix h, b, d and strictly decrease e . In every case the lexicographically leading changed coordinate strictly decreases and no earlier coordinate increases, so $\mu(T') < \mu(T)$. As $<$ on $\mathbb{N}^4$ is well-founded, the system terminates. $\square$

3 Local confluence

Proposition 5 (local confluence). If $T \rightsquigarrow T_1$ and $T \rightsquigarrow T_2$ by two (possibly different) rules, then there is a tree U with $T_1 \rightsquigarrow^* U$ and $T_2 \rightsquigarrow^* U$.

Proof (critical-pair analysis). Two redexes either act on vertex-disjoint parts of T or overlap.

Disjoint redexes commute. If the two applied rules touch disjoint edge sets, applying one does not disturb the other’s redex, so T_1 and T_2 both reduce to the tree U obtained by applying both; this closes the diamond in one step each. This covers all pairs of rules applied at different hubs/legs.

Overlapping redexes (critical pairs). Overlap is possible only when both redexes involve a common center. Enumerate by the leading rule:

- **(H)/(H)** at a shared hub: merging hub a with b versus a with c . Both results still contain the third hub, and a further **(H)** on each side merges it, yielding the same single-hub tree U (hub merges are associative on the finite hub set). Arm rebalancing (part of **(H)**) is a function of the final arm multiset, so it agrees.
- **(B)/(B)** at a shared hub, different branches: the two branch replacements are independent (each replaces one branch by an arm), so they commute; U has both replaced.
- **(B)/(K)**: a Kelmans compression moving a pendant unit past a hub can relocate a branch that **(B)** would replace. Apply **(B)** to the relocated branch afterward: T_2 (post-**(K)**) still has the non-arm branch, now at the target hub, and **(B)** replaces it there; T_1 (post-**(B)**) has an arm that **(K)** then compresses. Both reach the tree with the branch turned into an arm at the top hub.
- **(K)/(K)**: two compressions of different pendant units are independent moves on the backbone and commute; two compressions of the same unit toward the same top hub are cofinal (both continue to the top).
- **(L)/:** a leg-to-cherry move commutes with every other rule (it edits a leg's internal shape, disjoint from hub/branch/backbone structure), except when the leg is itself the branch a **(B)** move replaces — then **(B)** subsumes **(L)** and both sides reduce to the arm.

In each overlapping case the two one-step reducts reduce further to a common U . Hence the system is locally confluent. $\square$

Remark 6 (strict monotonicity of **(K)**, closed for all N ; superseded by the merge layer). [2026-08-15: this remark's uniform-load certificate is subsumed by the mixed-load merge layer — the exchange dichotomy and topped-up merge cover per-vertex loads, arbitrary capped environments, and de-loaded donors; see §??.] Local confluence above is purely combinatorial (it does not use A). Confluence gives a unique normal form as a tree; that this normal form maximizes A uses that every step is A -non-decreasing (Definition ??), which is R2-R6, and termination toward the star additionally needs **(K)** to be strictly increasing off the isomorphic locus. This strictness is now closed for all N . The Kelmans amplitude difference $\Phi = \pi' - \pi$ is bilinear on a box and hence $\geq$ its minimum corner; each of the four corner numerators, after the shift $d_a = 1 + u$, $d_b = 2 + v$, $c = 3 + s$ (with $u, v, s \geq 0$), has the form

$$N(u, v, s) = \kappa u + R(u, v, s), \quad \kappa > 0, \quad R \text{ a polynomial with all coefficients } \geq 0,$$

with $\kappa \in \{99, 945, 1080, 6480\}$ for the four corners (`psi_close.certify_corners_strict`, symbolic in d_a, d_b, c). Since $R \geq 0$, we get $N \geq \kappa u$, so $\Phi > 0$ whenever $u = d_a - 1 > 0$, i.e. at any non-endpoint backbone center; and $\Phi = 0$ forces $u = 0$ ($d_a = 1$, a backbone endpoint — the degenerate isomorphic recentering). A non-star backbone always has a center of backbone-degree ≥ 2 , so it admits a strictly π -increasing Kelmans move. Being symbolic in d_a, d_b, c , this holds for every N , not just a finite check.

4 Confluence and the maximizer

Theorem 7 (Newman). *A terminating, locally confluent rewriting system is confluent: every element has a unique normal form.*

Corollary 8 (unique normal form). *By Propositions ?? and ?? and Theorem ??, every tree T has a unique normal form $\text{NF}(T)$ under $\rightsquigarrow$. A tree is a normal form iff no rule applies: no non-arm branch (else (B)), a single hub (else (H)), no compressible displacement (else (K)), and all legs are cherries with arms balanced at five and the hub de-loaded (else (L)/rebalance). The unique such tree on the appropriate vertex count is the de-loaded single-hub five-cherry-bundle star S^* .*

Theorem 9 (R7, modulo R3). *For all large n , $\text{NF}(T) = S^*$ and $A(T) \leq A(S^*) = C_1$ with equality iff $T = S^*$; hence S^* maximizes π over n -vertex trees. This is unconditional given $\Phi \leq 1$ as a theorem (R3); the remaining ingredients — branch multiplicativity, R2, R4 (with its strictness, Remark ??), R5, R6 — are proved for all N .*

Proof. By Corollary ??, $T \rightsquigarrow^* S^*$. Each step is A -non-decreasing (Definition ??, using R2–R6), so $A(T) \leq A(S^*)$. The value $A(S^*) = C_1$ is the surface constant of rem:tie (single de-loaded hub), computed exactly. Equality forces every step to be non-strict, which by the strictness clauses of Definition ?? (and, for (K), the strict corner certificate of Remark ??) forces T to already be a normal form, i.e. $T = S^*$. $\square$

5 What remains

[Updated 2026-08-15.] The assembly program has advanced far beyond the earlier draft of this section. Machine-checked in Lean 4 (no `sorry`, no axioms, CI-verified): $\Phi \leq 1$ for the branch cavity model (`phi_le_one`); the full bridge from that model to $\text{per } L(T) / \prod \deg$ of the literal trees (the 2026-08-14 capstones); the merge layer — the 36-cell topped-up merge table, head-merge monotonicity, and `chain_to_normalForm`; the (L) classification layer; and the four shedding lemmas of the de-loading schedule. Proved by exact rational certificates (no floating point anywhere in a certificate path): the fixed- n rate bound $\pi(T) \leq \frac{4}{3}\rho_B^n$ for every tree; the slack-ledger dichotomy with rationally certified class floors; the amortized hub bound (critical profile family symbolic); the stuck-configuration dominations at two hubs (all sizes, symbolic tails plus a 442,800-case exact sweep) and for depth-2 hub stars (972 symbolic certificates); the de-loading schedule for $K \geq 40$ with the exact finite table below; and the interpolation lemma closing all sub-hub size vectors. An exhaustive check confirms the single-hub template family contains the true maximizer at every $n \in [8, 17]$.

The final theorem is being assembled as an honest conditional: a Lean capstone over *named hypotheses*, each carrying its certificate. Two items deserve emphasis:

Depth ≥ 3 genericity. The elimination of deep defected hub trees rests on the generic ledger threshold plus adversarial sampling; margins grow with depth and hub count in every probe, but this link is at sampling rigor — the one remaining probe-grade hypothesis.

Large exact sweeps. The finite dominations are exact rational computations, *decide*-shaped but large; the development bans `native_decide`, so these are cited as reproducible artifacts pending a formalization-policy decision.

Everything else — including the boundary caveat this note previously carried — is closed at theorem grade. Nothing here is a formal claim until the assembly lands and the whole development is independently refereed; the internal ledger accordingly still records `conjecture1_proved = False`.